

1. P      $\vee$    $\equiv$  $- 3.14.7$

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\VSC\files ng pyhton\Outputs NUMSOL> python --version
Python 3.14.7
PS D:\VSC\files ng pyhton\Outputs NUMSOL> 
```

2. P $\vee$ $\cap \equiv \emptyset / q$

The image shows a Visual Studio Code (VS Code) editor window. The editor has a dark theme. At the top, there is a tab for a file named `happy.py`. The code in the editor is a single line: `print("python is working")`. Below the editor, there is a terminal window. The terminal has a tab labeled `OUTPUT`. The terminal shows the command `python -u "d:\VSC\files ng pyhton\Outputs NUMSOL\happy.py"` being executed. The output of the command is `python is working`. Below the output, there is a message: `[Done] exited with code=0 in 0.081 seconds`.

```
happy.py X
happy.py
2 print("python is working")

PROBLEMS OUTPUT ... Filter (e.g., text, excludeText, t...) Code
[Running] python -u "d:\VSC\files ng pyhton\Outputs NUMSOL\happy.py"
python is working

[Done] exited with code=0 in 0.081 seconds
```


3 > 4. S —     // —  $\angle \emptyset \cap$ P  $\angle$  //   
  $\angle \cap$ .

```
lab00_check_Balobal.py X
lab00_check_Balobal.py > ...

22
23 # ---- 3. SciPy cross-check ----
24 area, _ = integrate.quad(np.sin, 0, np.pi)
25 print("integral of sin(x) from 0 to pi :", round(area, 12), "(exact value 2)")
26
27 # ---- 4. Matplotlib figure ----
28 os.makedirs("figures", exist_ok=True)
29 plt.rcParams.update({'figure.dpi': 150})
30
31 fig, ax = plt.subplots(figsize=(5.2, 3.2))
32 ax.plot(x, y, color="#b05a5a", lw=1.6, label="sin(x)")
33 ax.plot(x, np.cos(x), color="#85a800", lw=1.2, ls="--", label="cos(x)")
34 ax.set_xlabel("x (radians)")
35 ax.set_ylabel("value")
36 ax.set_title("Lab Assignment 0 - platform check")
37 ax.grid(True, alpha=0.3)
38 ax.legend()
39 fig.tight_layout()
40 fig.savefig("figures/lab00_check.png")
41
42 print("\nfigure written to figures/lab00_check.png")
43 print("PLATFORM READY.")
```

PROBLEMS OUTPUT ... Filter (e.g. text, !excludeText, t... Code

[Running] python -u "d:\VSC\files ng pyhton\Outputs NUMSOL\lab00_check_Balobal.py"

Python 3.14.7
 NumPy 2.5.2
 SciPy 1.18.0
 Matplotlib 3.11.1

mean of sin(x) over one period : 0.0
 integral of sin(x) from 0 to pi : 2.0 (exact value 2)

figure written to figures/lab00_check.png
 PLATFORM READY.

5. $G \equiv \begin{array}{c} \nearrow \nearrow \nearrow \\ \text{---} \end{array} \perp$
 $R \bigcirc \textcircled{P} \square \textcircled{S} \equiv \begin{array}{c} \nearrow \nearrow \nearrow \\ \text{---} \end{array} \square \bigcirc \text{---} \triangle$

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README

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